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Friends Also Matter: Examining Friendship Adjustment Indices as Moderators of Anxious-Withdrawal and Trajectories of Change in Psychological Maladjustment

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Abstract

The present study evaluated whether three indices of friendship adjustment (mutual friendship involvement, friendship stability, friendship quality) are important, but overlooked, moderators of the impact of anxious-withdrawal on trajectories of psychological maladjustment during early adolescence. Participants included 271 young adolescents (51% boys; $M_{\text{age}} = 11.83$ years; 42% ethnic minority) who completed peer nomination and self-report measures at three waves in a longitudinal study. In line with hypotheses, growth curve analyses revealed that anxious-withdrawal predicted less negative psychological adjustment trajectories for friended young adolescents, albeit somewhat differently for boys and girls. Understanding the unique significance of friendships in the lives of anxious-withdrawn young adolescents could improve interventions as their friendships may be more successfully targeted than their group-level peer difficulties.

Keywords

Anxious-withdrawal; Friendship; Loneliness; Depression; Early Adolescence

Anxious-withdrawal (or the behavioral tendency to withdraw from and avoid peers due to fears, anxieties, and self-consciousness; also known as shyness, anxious-solitude) poses significant psychosocial risk for children and adolescents (Rubin, Coplan, & Bowker, 2009). This risk is especially pronounced in the prediction of psychological difficulties, such as depression and loneliness (e.g., Karevold, Ystrom, Coplan, Sanson, & Mathiesen, 2012; Rubin, Chen, McDougall, Bowker, & McKinnon, 1995). Studies examining trajectories of depressive symptoms associated with anxious-withdrawal indicate that one type of peer experience, specially *group*-level peer experiences (such as peer exclusion), predict depressive symptoms for highly anxious-withdrawn youth (Gazelle & Rudolph, 2004; Gazelle & Ladd, 2003). Other types of peer experiences, however, such as those at the *dyadic*-level of social complexity, that may also be linked to longitudinal changes in the trajectories of psychological maladjustment associated with anxious-withdrawal have been largely overlooked. From a prevention and intervention perspective, group-level peer experiences are oftentimes very challenging to lever and change, but dyadic-level peer experiences, such as friendships, can be more successfully targeted (Asher, Parker, &

Walker, 1996; Hoza, Mrug, Pelham, Greiner, & Gnagy, 2003; Mikami, 2010). Accordingly, to address this limitation in the literature, this study investigates anxious-withdrawal and trajectories of change in psychological maladjustment, and whether three indices of friendship adjustment theoretically- and empirically-implicated to offset psychological difficulties (mutual friendship involvement, friendship stability, friendship quality; Hartup, 1996; Rubin, Bukowski, & Bowker, 2015; Sullivan, 1953), particularly during the early adolescent developmental period (10–14 years), serve as moderators. We also seek to clarify the role of gender in these models.

Theoretical Framework

Theoretical models of anxious-withdrawal posit that the psychological problems experienced by anxiously-withdrawn youth are rooted in their temperamentally-based fears and anxieties (which, in part, motivate their withdrawn behaviors) *and* their poor relationships (which are presumed to reinforce fears, anxieties, and insecurities; Rubin, 1993; Rubin et al., 2009; Rubin, LeMare, & Lollis, 1990; Rubin, Stewart, & Coplan, 1995). Most of the empirical work on poor *peer* relationships focuses on those at the group-level of social complexity (Rubin et al., 2015). Empirical evidence demonstrates that most peers judge anxious-withdrawn behaviors as atypical and non-normative (Coplan, Girardi, Findlay, & Frohlick, 2007). Such negative cognitive appraisals of anxious-withdrawal lead to peer rejection, peer exclusion, and peer victimization, which in turn, predict psychological difficulties (e.g., depression, loneliness) for anxiously-withdrawn children and young adolescents (Gazelle & Ladd, 2003; Gazelle & Rudolph, 2004; Boivin, Hymel, & Bukowski, 1995; Parkhurst & Asher, 1992). For example, Gazelle & Rudolph (2004) found that anxious-withdrawal predicted increases in trajectories of depressive symptoms among young adolescents who experienced high levels of peer exclusion, but that these associations were attenuated in the context of low peer exclusion. Related research indicates that anxious-withdrawn youth who experience low levels of peer exclusion and victimization become less socially fearful and avoidant over time (Booth-LaForce, Oh, Kennedy, Rubin, Rose-Krasnor, & Laursen, 2012; Shell, Gazelle, & Faldowski, 2014). Thus, exposure to negative group-level peer contexts appears to strengthen the negative psychological concomitants of anxious-withdrawn dispositional styles while less negative group-level peer contexts ameliorate such difficulties. These findings are consistent with not only etiological models of anxious-withdrawal but also child-x-environment and diathesis-stress models that emphasize the dynamic interplay between individual vulnerabilities and environmental risk factors in the prediction of risk and adaptation (Joiner, Heatherton, Rudd, & Schmidt, 1997; Magnusson & Cairns, 1996; Magnusson & Stattin, 2006),

The most recent etiological models of anxious-withdrawal (see Coplan & Bowker, 2014; Rubin et al., 2009) specify that poor peer experiences at *both* the group- and dyadic-levels of social complexity can explain why some anxious-withdrawn youth come to experience more psychological maladjustment than others. At the dyadic-level, researchers argue that there are at least three ways in which youth might succeed or struggle with friends or three different indices of friendship-adjustment: (1) forming mutual friendships; (2) maintaining their friendships; and (3) developing high-quality positive friendships (Rubin et al., 2015). Decades of research show that youth who are unable to form mutual friendships, and

especially those that are lasting and high-quality, are more likely to struggle psychologically (in terms of depressive symptoms, loneliness, self-esteem) than those who are able to form friendships (e.g., Nangle, Erdley, Newman, Mason, & Carpenter, 2003; Parker & Asher, 1993; Rubin et al., 2015; Sullivan, 1953). These findings emerge after controlling for group-level peer difficulties, suggesting that group- and dyadic-level peer experiences are not redundant predictors of adjustment (Rubin et al., 2015). Of course, the specific processes by which different types of friendship-adjustment impact youth likely differ. For instance, being without any mutual friend may make youth appear more vulnerable to would-be bullies and feel socially insecure, while lacking a high quality friendship and its offerings of intimate exchange and support may make it especially challenging for youth to cope with peer stress (Demir & Urberg, 2004; Rubin, Wojslawowicz et al., 2006; Sullivan, 1953). Unstable friendship involvement may uniquely foster a sense of inconsistent belongingness amongst peers (Poulin & Chan, 2010). Although friendships matter for children, they become increasingly intimate and influential on adjustment outcomes during the early adolescent developmental period (Sullivan, 1953). Of note, it is also during early adolescence when youths' tendencies to avoid or approach peers appear to become increasingly influential on their peer experiences (Gazelle & Rudolph, 2004).

Building off the recent models of anxious-withdrawal, the extensive friendship literature, and child-x-environment and diathesis-stress models (Joiner et al., 1997; Magnusson & Stattin, 2006; Rubin et al., 2009), the current study examines the potential contribution of friendship to the psychological risk associated with anxious-withdrawal by testing whether three specific indices of friendship-adjustment (mutual friendship involvement, friendship stability, friendship quality) influence the development of depressive symptoms and loneliness associated with anxious-withdrawal. Multi-dimensional assessments of friendship-adjustment have long been discussed as necessary to capture the many complexities in children's and adolescents' friendship experiences (Bukowski & Hoza, 1989; Hartup, 1996; Nagle et al., 2003; Rubin et al., 2015), but are rarely utilized, particularly in anxious-withdrawal research.

Anxious-Withdrawal and Friendship

The important focus on anxious-withdrawn youths' poor group-level peer experiences has resulted in relatively less attention to their dyadic-level peer experiences or their friendships. However, some available evidence indicates that anxious-withdrawn children and adolescents are as likely as their non-withdrawn counterparts to have at least one mutual and stable friendship (Ladd & Burgess, 1999; Ladd et al., 2011; Oh et al., 2008; Rubin, Wojslawowicz et al., 2006; Schneider, 1999). This may be because they find and form friendships with similarly withdrawn and excluded/victimized peers (Rubin, Wojslawowicz, et al., 2006). There is some indication though that these friendships are poor in terms of relationship quality (e.g., low levels of fun, companionship, high levels of conflict; Menzer et al., 2012; Ponti & Tani, 2015; Rubin, Wojslawowicz et al., 2006; Schneider, 2009). Anxious-withdrawn youths' anxieties and fears may interfere with their abilities to provide social support to their friends, and engage in intimate disclosure, a hallmark of friendships beginning in early adolescence (Rubin et al., 2015). Anxious-withdrawn young adolescents also have a less sophisticated understanding of friendship than their non-withdrawn

counterparts such that they prioritize receiving help over intimate exchange with their close friends (Schneider & Tessier, 2007). As such, it may be that they perceive their friendships negatively when their friends do not or cannot provide them with their desired amount of help and support.

The question of central interest in this study is whether such friendship experiences “matter” or can off-set psychological risks with offerings of social support, security and companionship. The focus of the aforementioned research was on *patterns* of friendship experiences. However, these studies did show that there is significant variability in anxious-withdrawn youths’ friendship experiences (likely due to variability in anxious-withdrawn youths’ behaviors and how they received by peers; Gazelle, 2008), and one line of research examined how variability in anxious-withdrawn youths’ friendships can influence them *socially* (Bowker & Spencer, 2010; Ladd et al., 2011; Rubin, Wojslawowicz et al., 2006). These studies showed that highly anxious-withdrawn young adolescents with *mutual* friends are rated by peers as more sociable and popular than those without mutual friends (Rubin et al., 2006), and that highly anxious-withdrawn boys are less victimized when they have mutual mixed-grade friendships (i.e., same-school, different-grade friendships; Bowker & Spencer, 2010). Moreover, a similar protective effect with regard to peer victimization emerged for involvement in *stable* mutual friendships (Ladd et al., 2011). These findings are complemented by research indicating that the lack of mutual best friendships and friendship *instability* predict increasing levels of anxious-withdrawal during early adolescence (Oh et al., 2008).

Absent from these studies is attention to *psychological* outcomes. Surprisingly, we could locate only three relevant studies that considered any psychological outcome. One was a cross-sectional study by Ponti and Tani (2015) that showed that the association between anxious-withdrawal and a measure of generalized internalizing distress is attenuated for young adolescents who report high quality friendships. Friendship quantity was not a significant moderator in this study. In another study of late adolescents, Vanhalst, Luyckx and Goossens (2014) found that friendship quality (but not quantity) mediated (but did not moderate) the concurrent associations between anxious-withdrawal and loneliness. Finally, in the only relevant longitudinal study, Bukowski, Laursen, and Hoza (2010) found that socially withdrawn young children who are also excluded by the larger peer group are at risk for increasing depressed affect, but that having mutual friends helped to minimize the risk.

Findings from these studies offer preliminary evidence that positive friendship experiences help to alleviate some of anxious-withdrawn youths’ psychological difficulties. Positive friendship experiences – for anxiously-withdrawn youth who otherwise struggle with peers – may provide unique relationship contexts that: (1) diminish insecurities and social fears through companionship, “consensual validation of personal worth” (Sullivan, 1953, p. 251), and reciprocated affection (*mutual friendship involvement*); (2) promote a sense of consistent belongingness (*friendship stability*); and (3) offer emotional and social support (*friendship quality*). But, of these studies, all were restricted to one (Bukowski et al., 2010) or two (Ponti & Tani, 2015; Vanhalst et al., 2014) indices of friendship-adjustment, and none considered friendship stability. Further, two were not longitudinal, and thus were limited in their ability to offer insight into developmental change. The one study that was longitudinal

focused on children (Grades 3–5 at the start of the study) and included only one psychological adjustment outcome (depressed affect, assessed by a peer nomination item for “sad”; Bukowski et al., 2010). The Bukowski et al (2010) study also focused on social withdrawal indexed by the peer nomination items: “shy”, “rather play alone than with others”, and “feelings get hurt easily”. These assessment items may have resulted in identification of a different subtype of social withdrawal, as recent approach and avoidance models of social withdrawal and advances in research (Coplan, Ooi, & Nocita, 2015; Coplan & Bowker, 2014) suggest that anxious-withdrawn youth *do* desire to play or be with their peers (more so than being alone) while avoidant youth do not. This distinction between anxious-withdrawal and avoidance-withdrawal is an important one because research also indicates other differences between these two related but distinct types of social withdrawal (e.g., avoidant youth actively dislike peer interaction while anxious-withdrawn youth do not; Coplan, Rose-Krasnor, Weeks, & Kingsbury, 2013; Coplan, Wilson, Frohlick, & Zelenski, 2006). Thus, there is a clear need to build on this earlier research by specifically examining anxious-withdrawal and friendship in relation to trajectories of psychological adjustment, with a focus on early adolescence (Rubin et al., 2015; Sullivan, 1953). Including additional indices of friendship adjustment and psychological adjustment outcomes (depressive symptoms and *loneliness*, which is commonly associated with both anxious-withdrawal and peer difficulties; Rubin et al., 2015) also has the potential to reveal new ways in which friendship is relevant to anxious-withdrawn youths’ psychological health on a broader scale than addressed in past research.

Anxious-Withdrawal and Gender

Gender is another potential moderator. There do not appear to be any gender differences in the likelihood that anxious-withdrawn boys and girls have mutual and stable friendships (Ladd et al., 2011; Rubin, Wojslawowicz et al., 2006). Findings regarding friendship quality, however, are mixed. Although some findings suggest that anxious-withdrawal is associated negatively with friendship quality for girls but not boys (Menzer et al., 2012), other studies indicate that the link does not vary by gender (Ponti & Tani, 2015; Rubin, Wojslawowicz et al., 2006; Schneider, 2009). It has been suggested that, while anxious-withdrawn boys may suffer more psychologically and with peers (because their behavior is judged as conflicting more strongly with male gender expectations for assertiveness and initiative), the psychological well-being of anxious-withdrawn girls may be impacted more so by negative peer experiences (Boivin et al., 1995; Bowker & Raja, 2011; Doey, Coplan, & Kingsbury, 2014). For instance, Gazelle and Rudolph (2004) found that highly anxious-withdrawn girls who were highly excluded displayed initial and relatively sustained elevations in depressive symptoms, whereas their male counterparts showed no initial elevation, but a subsequent increase in depressive symptoms across one year. Anxious-withdrawn young adolescent girls may be more sensitive and reactive to interpersonal difficulties than boys due to heightened concerns over interpersonal belonging during early adolescence relative to their male counterparts (Gazelle & Rudolph, 2004; Rose & Rudolph, 2006). Therefore, anxious-withdrawal might also be associated with higher initial levels and greater increases in depressive symptoms and loneliness trajectories in the context of poor *friendship*

experiences (friendlessness, friendship instability, and low friendship quality) for girls than boys.

Overview of the Present Research

Rubin and colleagues have advocated in several recent reviews of the social contributors to anxious-withdrawal and associated psychological difficulties that we would be mistaken to continue to focus future research solely on *group*-level peer experiences, particularly when studying anxious-withdrawal during early adolescence (Rubin et al., 2015; Rubin et al., 2009). Identifying varied peer experiences that enhance the psychological difficulties associated with anxious-withdrawal has important theoretical implications for understanding how and why anxious-withdrawal is such a strong risk factor for developmental psychopathology and applied implications for targets of intervention with anxious-withdrawn youth who are struggling. Accordingly, in the present study, we seek to advance theory and research by testing the hypothesis that positive friendship experiences, in the form of mutual friendship involvement, friendship stability, and positive friendship quality, will moderate the impact of anxious-withdrawal on trajectories of psychological maladjustment during early adolescence. Protective effects for each friendship experience were expected (although as discussed previously, vis-à-vis different processes; Poulin & Chan, 2010; Rubin et al., 2006; Sullivan, 1953). This hypothesis is evaluated using three time points of self- and peer-report data from a large longitudinal sample of Grade 6 young adolescents (who transition into Grade 7). Gender is also evaluated as a moderator, with the expectation that anxious-withdrawn girls would be impacted more so by their friendship experiences than boys.

Method

Participants

Participants were 271 young adolescents (51% boys; $M_{\text{age}} = 11.83$ years, $SD = 0.49$, at the start of the study, Time 1), from two suburban public middle schools in Buffalo, NY, in the United States, who were part of a longitudinal study on short-term changes in peer relationships during early adolescence (references removed for blind review). All sixth-grade students at the two schools were recruited for participation. Letters describing the study and parental consent and assent forms were sent home with all students; written parental permission and adolescent assent were obtained for all participants, with a consent rate of 70%. Fifty-eight percent of participants self-identified as Caucasian, 21% as African-American, 12% as Biracial, 4% as Hispanic, 3% as Arabic, 1% as Native American, and 1% as Asian, which is similar to the ethnic composition of the schools that the participants attended. Data on socioeconomic status was not collected from the students, but publicly available information indicates that the median income for the families in the participating school districts ranged between \$44,969 and \$54,098. Data of interest to this study were collected on three occasions (that were each approximately 3 to 3.5 months apart): Time 1 (T1: Grade 6, early February), Time 2 (T2: Grade 6, end of May/early June), and Time 3 (T3: Grade 7, early September).

Procedure

At each time point, participants completed measures in group-format settings such as the school classrooms or the cafeteria. Each administration session lasted 30 to 45 minutes. Participating students were told that their answers were confidential and that they could choose to stop completing the questionnaires at any time. Various incentives were used to ensure the return of forms. For example, each student, regardless of their decision to participate, received a university t-shirt and was entered into a raffle for a \$20 gift certificate. The three homerooms that returned their consent forms first, irrespective of their participation rate, were also rewarded with a pizza party. Retention rates in this study were very high; only 10 students dropped out of the study (due to leaving the school) between T1 and T3. Analyses indicated no differences in the study variables for participants who completed measures at all three time points relative to participants who completed measures only at one or two time points ($ps > .05$).

Measures

Anxious-withdrawal (T1)—Anxious-withdrawal was assessed with four peer nomination items used in past research (“very shy”; “doesn’t talk much or talks quietly”; “gets nervous about group discussion”; “hardly ever starts up a conversation”; Oh et al., 2008; Rubin, Wojslawowicz et al., 2006). Nominations for these items were unlimited, with same- and other-sex nominations permitted (for any peers in their grade at their school). Consistent with past research (e.g., Bowker & Raja, 2011; Bowker & Spencer, 2010; Oh et al., 2008), no class lists were provided; instead, participants were instructed to write the names of peers for each item, and only nominations for participating peers were utilized. Consistent with past growth curve modeling research (e.g., Spangler Avant, Gazelle, & Faldowski, 2011), the number of nominations received for each item was first summed, and then proportionalized by dividing the total number of nominations received by an exposure variable. The exposure variable was defined by the algorithm $n \times r$, which reflects the total number of students completing peer nominations at each school (n) multiplied by the number of items (in this case four) used to create the composite (r) (Spangler Avant et al., 2011). Thus, the final adjusted anxious-withdrawal variable was a proportion score reflecting the percentage of nominations received out of the total number of possible nominations (indexed by the total number of students completing peer nominations *and* the number of independent chances each youth had to receive a nomination; Spangler Avant et al., 2011). Internal consistency of this mean scale in this study was good ($\alpha = .82$).

Mutual friendship involvement and stability (T1 and T3)—Participants wrote the names of their same-sex “very best friend” and “second best friend,” and three same- or other-sex “good” friends from their grade and school (Bukowski, Hoza, & Boivin, 1994; Rubin, Wojslawowicz et al., 2006). Friendships were considered mutual if the nominations were reciprocated either as a “best” or “good” friend. In the present study, at both time points, approximately 70% of participants had at least one mutual friend, similar to percentages reported in prior research (e.g., Parker & Asher, 1993; Rose, 2002). Also similar to prior studies, girls were more likely to have a mutual friendship than boys at T1, $\chi^2(1) = 16.92, p = .001, \eta^2 = 0.27$. Of the participants with a mutual friendship at T1, 70% had at least one “best” or “good” friendship that was stable from T1 to T3 (determined by *mutual*

friendship nominations at both time points, regardless of whether the nominations were *consistently* for “best” or “good” friends). If a young adolescent had more than one mutual friendship at T1, he or she was considered to have a stable mutual friend if any of these friendships were maintained at T3 (for a similar procedure, see McDonald et al., 2011; Rubin, Wojslawowicz et al., 2006). Boys (67%) and girls (73%) did not significantly differ from each other in the likelihood of having a stable mutual friendship, $\chi^2(1) = 0.28, p = .60, \eta^2 = .05$.

Friendship quality (T1)—The 23-item *Friendship Quality Scale* (FQS; Bukowski et al., 1994) was used to assess the quality of young adolescents’ relationships with their best friends. The FQS includes five subscales: companionship (4-items; e.g., “My friend and I spend all our free time together”), closeness (5-items; e.g., “If my friend had to move away, I would miss him/her”), help (5-items; e.g., “My friend helps me when I am having trouble with something”), security (5-items; e.g., “If my friend or I do something that the other doesn’t like, we can make up easily”), and conflict (4-items; e.g., “My friend and I argue a lot”). Participants were asked to rate how true each item was for their best friend relationship (i.e., the relationship with their very best friend identified in the friendship nominations portion of the survey packet; 72% of which were mutual friendships) on a 7-point Likert scale (1 = not true; 7 = really true). In the present study, a total positive friendship quality score was calculated by reverse-scoring the conflict scale and averaging all five subscales (Markiewicz, Doyle, & Brendgen, 2001). Internal consistency of the mean scale was excellent in this study ($\alpha = .91$).

Loneliness and depressive symptoms (T1-3)—Loneliness was assessed with the 16-item *Loneliness and Social Dissatisfaction Questionnaire* (LSDQ; Asher, Hymel & Renshaw, 1984; “It’s hard for me to make friends”). At each time point, participants were asked to rate each item on a 5-point Likert scale (1 = *not at all true*; 5 = *really true*). Depressive symptoms were assessed with the 10-item *Children’s Depressive Inventory: Short Version* (CDI:S; Kovacs, 1992; e.g., “I am sad all of the time”). For each item (at each time point), participants were presented with three statements and asked to pick *one* statement that best describes their feelings in the past two weeks (e.g., “I do not feel alone”, “I feel alone many times”, or “I feel alone all the time.”; coded as 1, 2, or 3). Mean scores were calculated with higher scores reflecting greater levels of loneliness and depressive symptoms. Items are typically summed for the CDI, but mean scores were computed for the CDI in this study, consistent with other studies that utilized the 10-item scale (e.g., Priess, Lindberg, & Hyde, 2009), to avoid confusion between the 10-item and 27-item forms of the CDI. The mean of the *sum* scores of the 10 CDI items in this study was 2.91, 2.87, and 2.91 at T1-3. Internal consistencies for these scales were acceptable across all three time points (loneliness, T1-T3: $\alpha = .92, .91, .93$; depressive symptoms, T1-T3: $\alpha = .73, .76, .76$).

Results

Descriptive Statistics and Correlations

Table 1 summarizes means, standard deviations and inter-correlations among the study measures, separately for boys and girls. T1 anxious-withdrawal was related positively to

loneliness and depressive symptoms at each time point for girls (but not boys). For both boys and girls, however, anxious-withdrawal was not associated with friendship quality, mutual friendship involvement, or friendship stability. Also consistent for boys and girls, depression and loneliness evidenced strong stability and moderate associations across time. Independent samples *t* tests explored mean-level gender differences. Relative to boys, girls were rated as more anxious-withdrawn, $t(269) = -2.57, p = .011, d = -0.19$, and rated their friendships as higher in relationship quality, $t(207) = -7.75, p = .001, d = -0.70$, but did not differ significantly on the other study variables.

Examining Friendship Indices as Moderators between Anxious-Withdrawal and Psychological Maladjustment Trajectories

To account for missing data and to examine for both individual- and group-level change, the study hypotheses were evaluated with growth-curve analyses using Hierarchical Linear Modeling (HLM; Raudenbush & Bryk, 2002). Separate models were built for loneliness and depressive symptoms. Separate conditional models were also specified for each index of friendship due to high degree of overlap between the friendship variables and concerns about multicollinearity. All independent variables were grand-mean centered, and the categorical variables of friendship involvement (-1 = no mutual friend; $+1$ = at least one mutual friend), friendship stability (-1 = no stable friend; $+1$ = at least one stable friend), and gender (-1 = boys; $+1$ = girls) were effect coded. In HLM models described below, negative parameter coefficients for dichotomous variables indicate higher levels of the criterion for groups coded -1 , and positive coefficients indicate higher levels of the criterion for groups coded $+1$. The loneliness and depression variables were skewed; transformations were applied but results are presented with untransformed data due to similar results with transformed data. Finally, individual-level change in loneliness and depressive symptoms was modeled as a function of time where Time=0 at T1, Time=1 at T2, and Time=2 at T3. In all growth curve models, only linear changes in loneliness and depression were considered because: (1) more than three time points are considered ideal for examining quadratic effects; (2) findings from recent longitudinal work suggest that linear models may offer a more parsimonious explanation for developmental trends in both loneliness and depression relative to non-linear models (Ladd & Ettekal, 2013; Qualter, Brown, Munn, & Rotenberg, 2010); and (3) past research indicates that anxious-withdrawal and group-level peer experiences *interact* to predict of linear changes in depression (Gazelle & Ladd, 2003; Gazelle & Rudolph, 2004). In the sections that follow, only significant findings are discussed.

Individual-level variability in depressive symptoms and loneliness—First, Level 1 analyses examined the extent of *individual* variation in each of the psychological outcomes (and whether conditional models were warranted). Results of these fully unconditional models revealed significant individual variation in initial levels (e.g., the intercept) of loneliness and depressive symptoms as well as the rate of change across time (e.g., the slope; see random effects in Table 2). The intercept and slope for loneliness and depressive symptoms were relatively stable on average, consistent with observed trends for internalizing problems during early adolescence reported by other researchers (e.g., depressive symptoms; Gazelle & Rudolph, 2004).

Impact of anxious-withdrawal and gender on depressive symptoms and

loneliness—We then evaluated partially conditional models that specified T1 anxious-withdrawal, gender, and their interaction as time-invariant T1 predictors at Level 2 for each outcome (Table 2). Anxious-withdrawal was associated with initially higher loneliness and predicted a significant linear increase in loneliness. Boys reported higher initial levels of loneliness compared to girls.

Impact of anxious-withdrawal, gender, and friendship on depressive

symptoms and loneliness—Six fully conditional models (three for each outcome) were next specified that included main effects of anxious-withdrawal, one of the friendship variables, and gender, as well as the two- and three-way interactions as time-invariant T1 predictors at Level 2.

As evident in Table 3, not having a T1 mutual friend predicted higher initial levels of loneliness. A significant two-way interaction between anxious-withdrawal and mutual friendship for the intercept also emerged, with follow-up analyses indicating that anxious-withdrawal was associated with higher levels of loneliness (relative to the mean) for those without a T1 friend ($\beta = 0.33, p < .001$), but not for those with a friend ($\beta = 0.02, p = .91$). A significant interaction involving anxious-withdrawal and gender was also found, with simple slope analyses indicating that anxious-withdrawal was not related to T1 loneliness for boys ($\beta = 0.03, p = .07$) but was related to T1 loneliness for girls ($\beta = 0.07, p = .001$). Higher friendship quality was associated with initially lower loneliness relative to the mean (coefficient = $-0.49, p = .001$).

As seen in Table 3, the absence of a T1 friend was associated with higher initial levels of depressive symptoms. Significant three-way interactions between anxious-withdrawal, mutual friendship, and gender predicting the intercept and the slope were found. Simple slope analyses revealed that for boys, anxious-withdrawal was not associated with T1 depressive symptoms, irrespective of their T1 friendship involvement (friendless boys: $\beta = -0.06, p = .47$; friended boys: $\beta = 0.09, p = .47$). However, anxious-withdrawal predicted linear *increases* in depressive symptoms for friendless boys ($\beta = 0.06, p = .02$), but linear *decreases* for friended boys ($\beta = -0.09, p = .006$; see Figure 1). For friendless girls, anxious-withdrawal was associated with higher initial levels of depressive symptoms ($\beta = 0.13, p < .001$) and predicted steady, albeit less steep, increases in depressive symptoms relative to the trajectories observed for boys ($\beta = 0.03, p < .001$). Anxious-withdrawal was not associated with initial levels of depressive symptoms ($\beta = -0.11, p = .20$) or significant changes in depressive symptoms across time ($\beta = 0.06, p = .16$) for friended girls (see Figure 2). Finally, higher friendship quality was associated with lower initial levels of depressive symptoms (relative to the mean; coefficient = $-0.12, p = .01$), and for both outcomes, there were no significant effects with friendship stability (additional output for models with friendship quality and stability, as well as the model equations are available from the first author by request). Also of note, the inclusion of peer rejection (assessed by peer nominations of “likes least”) as a covariate did not affect the pattern of results in any substantive way, and age and the categorical school variable were not related significantly to any of the study variables ($r_s = .08-.14, p_s > .05$).

Discussion

Children and adolescents' anxious-withdrawn behavioral tendencies predict a broad spectrum of psychological difficulties. However, little progress has been made in identifying different *peer* sources of variability in anxious-withdrawn youths' psychological functioning. This study responds to calls for increased investigation of anxious-withdrawn youth's *friendship* experiences that are likely to influence their adjustment in meaningful ways. We examined potential moderator effects of three aspects of friendship adjustment on trajectories of depressive symptoms and loneliness related to anxious-withdrawal across an 8-month period during early adolescence, the developmental period during which friendships and avoidant behavioral tendencies both become increasingly influential on adjustment outcomes (Rubin et al., 2015). Current complex etiological models of anxious-withdrawal (Rubin et al., 2009) assume that both group- and dyadic-level peer experiences impact psychological adjustment, but the latter have been largely overlooked.

Consistent with prior studies, correlations showed that initial anxious-withdrawal was associated with greater loneliness and depression at each time point. Partially conditional models showed that anxious-withdrawal was associated with higher initial levels and increases in loneliness over time. More importantly, growth curve analyses revealed moderator effects: Mutual friendship involvement emerged as a moderator of the psychological trajectories associated with anxious-withdrawal, although the nature of the effects differed across the two outcomes, and for boys and girls. These results are compatible with child-x-environment and diathesis-stress models, and also fit well with existing knowledge that friendship involvement (or lack thereof) can alter the psychological well-being of young adolescents, and especially those at-risk, including avoidant-withdrawn children (Bukowski et al., 2010). Our findings are the first though to indicate that friendship experiences are also important for understanding the psychological trajectories of *anxious*-withdrawn young adolescents.

As noted, the moderator effects of mutual friendship involvement differed for loneliness and depression. Mutual friendship involvement functioned protectively for *all* anxious-withdrawn young adolescents with regard to initial levels of loneliness. This finding is novel as no previous studies in this research area included loneliness as an outcome. Anxious-withdrawn young adolescents who are able to form mutual friends may be able to feel socially secure and satisfied, despite their struggles with the larger peer group. On the other hand, anxious-withdrawn young adolescents who fail to form mutual friendships likely experience profound feelings of social dissatisfaction and despair due to a "double whammy" effect of social failure at both the group- and dyadic-levels of social complexity (Parker & Asher, 1993). Of course, the association between anxious-withdrawal and loneliness for friendless youth was significant but small, and therefore, it is possible that some anxious-withdrawn young adolescents without same-grade, same-school mutual friends have mutual friendships with peers outside of their same-grade and schools that help protect them from feelings of loneliness (Bowker & Spencer, 2010). Neighborhood friendships become more common and influential throughout adolescence (Rubin et al., 2015). Thus, it may be that assessments of friendships that are not limited by school setting reveal that anxious-withdrawn young adolescents' friendship networks are much larger and

more complex in terms of influence than available knowledge suggests. Nevertheless, the present results suggest that simply having one mutual same-grade and same-school friend can diminish feelings of *loneliness* for young adolescents with anxious-withdrawn behavioral tendencies.

The findings on depressive symptoms differed, yielding a more complex relation between anxious-withdrawal, friendship, and depression than was originally theorized. In our study, anxious-withdrawn friendless girls reported higher initial levels of depressive symptoms but we found that friendlessness placed both anxious-withdrawn young adolescent boys and girls on trajectories of increasingly high levels of depressive symptoms. Mutual friendship involvement mitigated these associations such that anxious-withdrawal was not related to significant changes in depressive symptoms for friended girls and predicted decreases in depressive symptoms for friended boys over time. Collectively, the pattern of findings provides evidence that friendlessness can intensify the depression risk associated with anxious-withdrawal (for both boys and girls) while friendedness can diminish the risk (for girls) or even promote psychological well-being (for boys).

The trajectory findings found in our study replicate the pattern reported by Bukowski and colleagues (2010) using an assessment of anxious-withdrawal (as opposed to avoidant-withdrawal) and depressive symptoms (rather than depressed affect) and an older age group. Moreover, like Bukowski et al. (2010), we found that the detrimental consequences of friendlessness on the depressive symptoms of anxious-withdrawn young adolescents persisted across a school grade transition when new friendships often develop (Rubin et al., 2015). As suggested by Gazelle and Ladd (2003) and Burgess et al. (2006), peer difficulties, in this case, friendlessness, may confirm anxious-withdrawn young adolescents' social fears, anxieties, internal blame tendencies, and feelings of helplessness, which in turn, transition into feelings of hopelessness and negative attributional biases that increase risk for depression. Forming new friendships (later in the school year or after a school grade transition), however, may not be sufficient to alter these patterns of thinking and feeling, which once developed, can be rather resistant to change (Gotlib, Lewinsohn, Seeley, Rohde, & Redner, 1993).

Effects of friendedness on anxious-withdrawn young adolescent boys' and girls' trajectories of depressive symptoms differed; in contrast, Bukowski et al (2010) did not find any significant gender differences in the protective effect of gender. This discrepancy is noteworthy and may be due to our older sample of young adolescents and gender differences in friendship experiences (Rose & Rudolph, 2006). Previous evidence indicates that girls are more cooperative and direct more prosocial behavior toward their friends; they also engage in more intimate disclosure with their friends. In contrast, boys tend to be more focused on companionship and fun-promoting activities with their friends. Many of these gender differences become more pronounced during adolescence (Rose & Rudolph, 2006). Both types of sex-linked behavioral styles have been posited to positively impact psychological adjustment, although negative effects of girls' behavioral styles have also been hypothesized (e.g., intimate disclosure could lead to co-rumination; Rose & Rudolph, 2006). It has also been found that social isolation (indexed by little time spent with same-sex friends and family members) during adolescence is more strongly associated with depression for boys

than girls (Larson, Raffaelli, Richards, & Ham, 1990). Given this evidence, it may be that anxious-withdrawn young adolescent boys and girls both benefit from having mutual friendships with their same-sex peers but that the sex-linked behavioral styles seen in boys' friendships are particularly beneficial for the psychological well-being of anxious-withdrawn boys. The emphasis on shared activities in anxious-withdrawn boys' friendships may not only help to diminish feelings of hopelessness and despair but could also help to promote behavioral approach patterns of peer engagement, and in turn, psychological health and well-being. Though further research will be required to test these hypotheses, our hypotheses are consistent with *behavioral activation* models of depression, which highlight the importance of activities to counter patterns of avoidance and withdrawal (e.g., Jacobson, Martell, & Dimidjian, 2001). Our findings also dovetail well with those from one study that suggested anxious-withdrawn boys might be more impacted by their friendship experiences than anxious-withdrawn girls (Bowker & Spencer, 2010).

We find it interesting that anxious-withdrawal was not related significantly to friendship stability or quality, and that friendship stability and quality did not modify the psychological adjustment trajectories associated with anxious-withdrawal. Although some earlier studies have suggested significant linkages between these constructs and shown that friendship quality and stability can be protective for anxious-withdrawn youth (e.g., Ladd et al., 2011; Menzer et al., 2012; Ponti & Tani, 2015; Rubin et al., 2006), findings from other studies have been consistent with our results (e.g., Fordham & Stevenson-Hinde, 1999). These contradictory findings may be the result of differences in the severity of anxious-withdrawn behaviors studied. Most past researchers focused on samples of youth whose anxious-withdrawal scores were at the top 33% (Ladd et al., 2011; Menzer et al., 2012; Ponti & Tani, 2015; Rubin, Wojslawowicz et al., 2006) or were clinically referred by mental health professionals (Schneider, 2009), and thus may represent more extreme exemplars of withdrawn youth relative to those included in this study. Anxious-withdrawal might be associated with lower friendship quality only among *extremely* anxiously-withdrawn youth, perhaps because highly anxious-withdrawn youth overly depend on their friends to deal with stressful social situations or because their high levels of anxieties and fears interfere more so with friendship processes and offerings of friendship provisions. Such closeness/dependency could be beneficial in the short-term (Ponti & Tani, 2015) and with regard to peer victimization (Ladd et al., 2011), although we would question whether it would be psychologically helpful in the long-term (Rose, 2002; Rose & Rudolph, 2006). Additional research is needed. Nevertheless, we believe that our results are noteworthy as they emphasize the importance of conceptualizing and measuring different types of friendship adjustment in studies of anxious-withdrawn youth. Further, they are significant in their suggestion that the long-term psychological significance of friendship for many anxious-withdrawn youth may have less to do with the length and intimacy of their relationships and more to do with simply having a companion by their side. It might be that there are some limitations on the extent to which having a stable and positive friendship can be psychologically helpful and "corrective" of anxieties, insecurities, and fears (Sullivan, 1953), beyond the benefits of simply having a friend.

One could also argue that having mutual friends means having supportive and stable friendships (for most young adolescents), and that the psychological importance of mutual

friendship involvement is somewhat confounded with that associated with friendship quality and stability (Bagwell, Newcomb, & Bukowski, 1998; Rubin et al., 2006). That said, we still believe that the impact of friendship quality and stability on the trajectories examined in this study is worthy of further inquiry due to limitations of our study. The current study relied on adolescents' self-reports of friendship quality, and future research may do well to include friend's reports of friendship quality. Friends do not always agree on the quality of their friendships (Brendgen, Little, & Krappmann, 2000), especially the friends of anxious-withdrawn youth who appear to view their friendships more negatively than do anxious-withdrawn youth (Rubin, Wojslawowicz et al., 2006). It may be that some anxious-withdrawn adolescents' friends have more accurate perceptions of the quality and interactions in the friendship, and that such perceptions do explain variability in anxious-withdrawal and psychological adjustment over time. In our study, not all students rated the quality of mutual best friendships, which tend to be higher in relationship quality. Thus, moderator effects of friendship quality might have also emerged if we had focused exclusively on those with mutual friendships.

There are several other significant study limitations and future research directions to consider. For example, investigators of social withdrawal have recently begun to move beyond global assessments of social withdrawal and towards a focus on not only anxious-withdrawal but also avoidant-withdrawal and unsociability-withdrawal, which differ from anxious-withdrawal in underlying motivation and associated psychosocial risk (Coplan et al., 2015; Rubin et al., 2009). The work on these latter types of social withdrawal also tends to focus on group-level peer experiences (for the one exception, see Bukowski et al., 2010). Our study focused on anxious-withdrawal, but additional studies that consider the psychological importance of friendship experiences to avoidant- and unsociable-withdrawn youth might make important contributions to our understanding of any subtype differences in the impact of friendship. Future research should also be expanded to consider friend characteristics. Although often discussed, few studies have considered the effects of having similarly-withdrawn friends. But, a *misery-loves-company* scenario may be important for understanding anxious-withdrawn youth's psychological trajectories (Rubin et al., 2009). Finally, we hope that future work will examine the *processes* responsible for the study effects and include longer longitudinal designs with more than three time points so that linear and non-linear effects can be best modeled (which may be especially important for loneliness, as suggested by the means in Table 1).

In closing, an assumption in recent theory and research on anxious-withdrawal is that the psychological difficulties experienced by anxious-withdrawal can be explained by difficulties at both the group- and dyadic-levels of social complexity. However, this assumption has not been examined in longitudinal research. Thus, our findings broaden understanding of peer influences on anxious-withdrawn youths' psychological difficulties by confirming that variations in anxious-withdrawn youths' loneliness and depressive symptoms can be explained by mutual friendship involvement and gender. The gender differences found in this study are novel and significant in their suggestion that the protective power of friendedness for anxious-withdrawn boys and girls may begin to differ somewhat during the early adolescent developmental period. Given the proven difficulties of changing youths' group-level peer experiences (Asher et al., 1996), our findings are

clinically significant as friendships (and especially friendship formation skills) can be more easily targeted in intervention efforts. We hope that our findings will lead to additional studies focused on different types of withdrawn boys and girls and the unique ways in which variability in different aspects of their friendship experiences matter.

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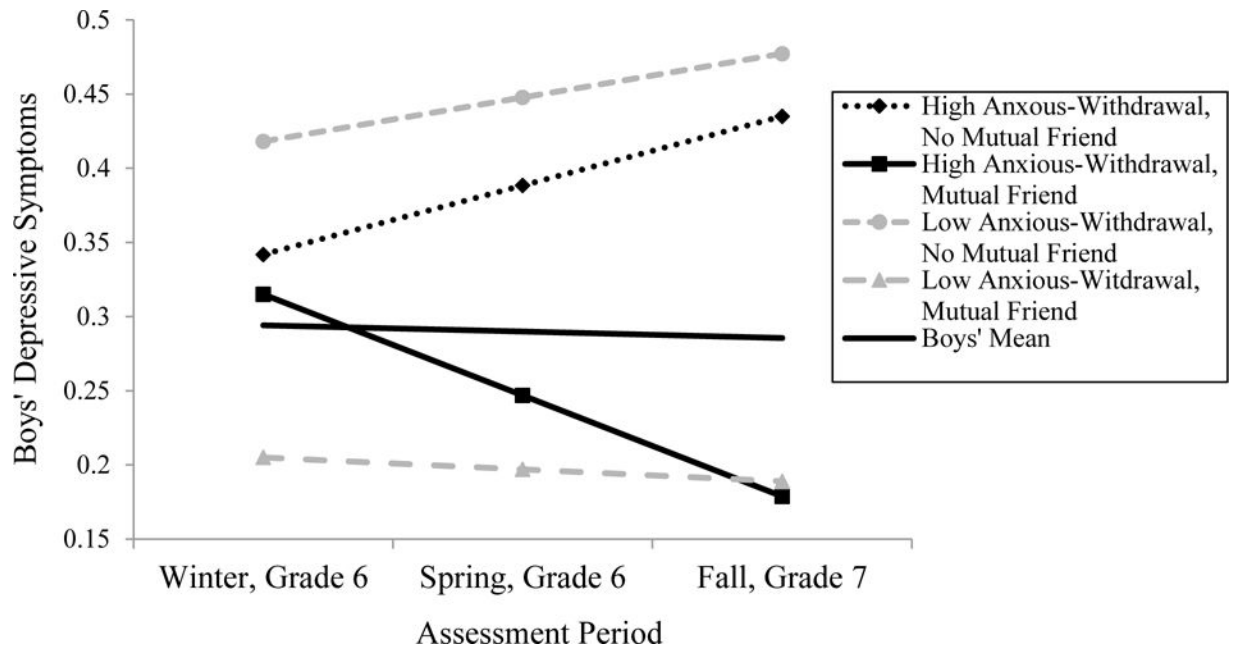


Figure 1.

Trajectories of depressive symptoms for anxious-withdrawn boys as a function of mutual friendship involvement in the winter of grade 6 (T1).

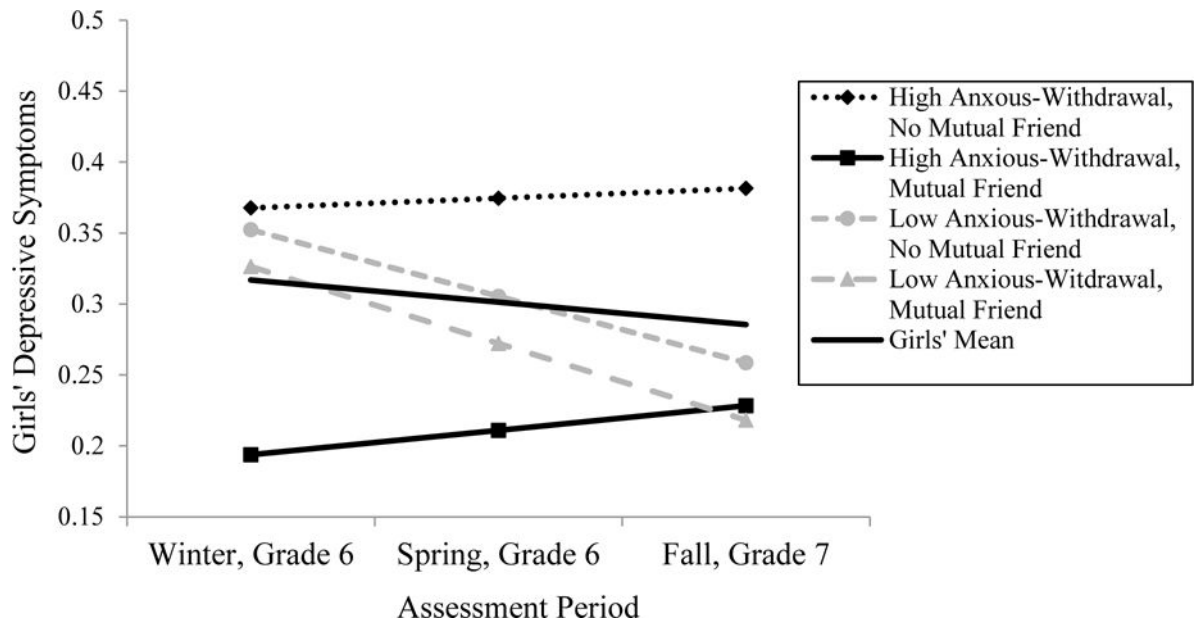


Figure 2.

Trajectories of depressive symptoms for anxious-withdrawn girls as a function of mutual friendship involvement in the winter of grade 6 (T1).

Table 1

Descriptive Statistics, Presented Separately for Boys and Girls.

	Girls <i>M</i> (<i>SD</i>)	Boys <i>M</i> (<i>SD</i>)	1	2	3	4	5	6	7	8	9	10
1. T1 Withdrawal	0.15 (1.05)	-0.14 (0.43)		.12	-.06	.22	.11	.17	.16	-.10	.17	.11
2. T1 Friend Quality	4.16 (.065)	3.49 (0.67)	.01		.25*	.20	-.45***	-.17	-.42***	-.33***	-.21	-.12
3. T1 Mutual Friend	0.84 (0.36)	0.59 (0.50)	-.16	.17		-	-.31**	-.20	-.17	-.14	-.21	-.23
4. Friend Stability	0.73 (0.45)	0.67 (0.47)	.05	-.09	-		-.05	.18	.09	.13	.07	.13
5. T1 Loneliness	1.54 (0.61)	1.73 (0.68)	.41***	-.37**	-.37***	.04		.51***	.78***	.31**	.70***	.54***
6. T1 Depression	0.29 (0.33)	0.31 (0.29)	.25*	-.20	-.29***	.24	.61***		.46***	.65***	.51***	.69***
7. T2 Loneliness	1.54 (0.58)	1.64 (0.68)	.52***	-.31**	-.37***	.07	.74***	.47***		.57***	.71***	.68***
8. T2 Depression	0.28 (0.29)	0.33 (0.36)	.34***	-.11	-.35***	.06	.39***	.65***	.58***		.31**	.79***
9. T3 Loneliness	1.60 (0.72)	1.68 (0.71)	.48***	-.36***	-.32**	.14	.65***	.51***	.75***	.56***		.67***
10. T3 Depression	0.28 (0.33)	0.33 (0.34)	.42***	-.09	-.30**	.12	.49***	.73***	.51***	.80***	.63***	

Note. Correlations for boys are presented above the diagonal and the correlations for girls are presented below the diagonal; T1 Mutual Friend: 0 = No mutual friends, 1 = At least one mutual friend; Friend Stability: 0 = No stable friendship, 1 = At least one stable friendship (from T1 to T3);

* $p < .05$;

** $p < .01$;

*** $p < .001$.

Table 2

Unconditional and Partially Conditional Growth Curve Models

Unconditional Model									
Fixed Effects	Loneliness				Depressive symptoms				
	Coefficient	SE	t-ratio		Coefficient	SE	t-ratio		
Model for initial status, π_{0i}									
Mean initial status, β_{00}	1.67***	0.05	36.37		0.32***	0.02		14.17	
Model for linear change, π_{1i}									
Mean linear change, β_{10}	-0.03	0.02	-1.15		-0.00	0.01		-0.15	
Random Effects									
	SD	RV	df	χ^2	SD	RV	df	χ^2	
Initial status, r_{0i}	0.60	0.36	179	981.60***	0.27	0.08	168	605.57***	
Growth rate, r_{1i}	0.19	0.04	179	315.06***	0.05	0.00	168	219.75**	
Level 1 error, e_{1i}	0.29	0.09			0.16	0.03			
Partially Conditional Model									
Fixed Effects	Loneliness				Depressive symptoms				
	Coefficient	SE	t-ratio		Coefficient	SE	t-ratio		
Model for initial status, π_{0i}									
Mean initial status, β_{00}	1.65***	0.04	36.83		0.31***	0.02		14.08	
Anxious-withdrawal, β_{01}	0.22*	0.11	2.04		0.06	0.04		1.49	
Gender, β_{02}	-0.11*	0.04	-2.41		-0.02	0.02		-1.07	
Aw-x-Gender, β_{03}	0.10	0.11	0.94		0.03	0.04		0.70	
Model for linear change, π_{1i}									
Mean linear change, β_{10}	-0.02	0.02	-1.02		-0.00	0.01		-0.42	
Anxious-withdrawal, β_{11}	0.06*	0.02	2.40		0.01	0.02		0.86	

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Unconditional Model									
Loneliness					Depressive symptoms				
Fixed Effects	Coefficient	SE	t-ratio	Coefficient	SE	t-ratio			
Gender, β_{12}	0.02	0.02	0.77	-0.01	0.01	0.01			-0.78
Aw-x-Gender, β_{13}	-0.04	0.02	-1.55	0.01	0.02	0.02			0.79
Level 1 pseudo-R ²	0.10			0.01					
Level 2 pseudo-R ²	0.11			0.02					
Loneliness					Depressive symptoms				
Random Effects	SD	RV	df	x ²	SD	RV	df	x ²	
Initial status, r_{0i}	0.57	0.32	176	834.57***	0.27	0.07	165	568.65***	
Growth rate, r_{1i}	0.19	0.04	176	306.30***	0.05	0.00	165	215.09**	
Level 1 error, e_i	0.29	0.09			0.16	0.03			

Note. N = 228 (Loneliness). N = 214 (Depressive symptoms).

* $p < .05$;
** $p < .01$;
*** $p < .001$;
RV = residual variance.

Table 3

Fully Conditional Models: Examining Mutual Friendship as a Moderator

Fixed Effects	Loneliness			Depressive symptoms				
	Coefficient	SE	t-ratio	Coefficient	SE	t-ratio		
Model for initial status, π_{0i}								
Mean initial status, β_{00}	1.69***	0.06	29.79	0.34***	0.03	12.15		
Anxious-withdrawal, β_{01}	0.11	0.06	1.75	0.01	0.04	0.30		
Gender, β_{02}	-0.03	0.06	-0.57	0.02	0.03	0.54		
MFI, β_{03}	-0.21***	0.06	-3.68	-0.08**	0.03	-2.81		
Aw-x-MFI, β_{04}	-0.17**	0.06	-2.74	-0.02	0.04	-0.52		
Aw-x-Gender, β_{05}	0.14*	0.06	2.30	-0.00	0.04	-0.02		
MFI-x-Gender, β_{06}	-0.00	0.06	-0.01	-0.02	0.03	-0.78		
Aw-x-MFI-x-Gender, β_{07}	0.08	0.06	1.32	-0.10*	0.04	-2.27		
Model for linear change, π_{1i}								
Mean linear change, β_{10}	-0.03	0.03	-0.94	-0.01	0.02	-0.49		
Anxious-withdrawal, β_{11}	0.09**	0.03	2.67	0.01	0.01	0.88		
Gender, β_{12}	0.01	0.03	0.24	-0.01	0.02	-0.76		
MFI, β_{13}	0.03	0.03	0.83	-0.00	0.02	-0.04		
Aw-x-MFI, β_{14}	0.02	0.03	0.59	-0.03*	0.01	-2.11		
Aw-x-Gender, β_{15}	-0.06	0.03	-1.71	0.03*	0.01	2.05		
MFI-x-Gender, β_{16}	-0.01	0.03	-0.36	0.01	0.02	0.58		
Aw-x-MFI-x-Gender, β_{17}	-0.01	0.03	-0.28	0.04***	0.01	3.07		
Level 1 pseudo-R ²	0.19			0.15				
Level 2 pseudo-R ²	0.23			0.16				
Random Effects								
	SD	RV	df	χ^2	SD	RV	df	χ^2
Initial status, τ_{0i}	0.49	0.24	155	571.40***	0.25	0.06	147	460.97***

Fixed Effects	Loneliness			Depressive symptoms		
	Coefficient	SE	t-ratio	Coefficient	SE	t-ratio
Growth rate, r_{ij}	0.19	0.03	155	0.07	0.00	209.09***
Level 1 error, e_{ij}	0.31	0.09		0.16	0.03	

Note. Δw refers to anxious-withdrawal. MFI refers to mutual friendship involvement. $N = 194$ (Loneliness), $N = 186$ (Depressive symptoms).

* $p < .05$;
** $p < .01$;
*** $p < .001$.